

Education

Worcester Polytechnic Institute (WPI)

Bachelor of Science in Computer Science, GPA 3.53/4.0

Worcester, MA

Aug 2022 - May 2026

Experience

ReVISit - AI Software Engineer, Remote

Aug 2025 to May 2026

- Built a full AI video analysis system integrated into revisit.dev, an open-source visualization platform cited in 13 peer-reviewed IEEE/CHI papers with 100,000+ total users
- Reduced manual video review time by 83%, from a mean of 217s to 35.6s per clip, by replacing manual scrubbing with a clickable confusion timeline spanning 6 event types
- Designed and benchmarked 4 AI pipelines: Gemini multimodal (summary quality at 2.67/3.0), GPT-4o frame-by-frame, LLaVA via Ollama for offline use
- Built a multi-signal fusion scoring formula combining audio, OCR, and scene signals into per-30-second confusion windows; added OCR grounding with Tesseract to boost confidence when confusion words appear in both transcript and on-screen text
- Implemented semantic search across all recordings using sentence-transformer embeddings (all-MiniLM-L6-v2, cosine similarity) and a cross-clip dashboard with task difficulty heatmaps and participant event density views
- Benchmarked GPT-4o at roughly 4.5x the per-clip cost of Gemini (\$0.018 vs \$0.004); results informed pipeline selection guidance in the evaluation framework

Sri Sangwan School - Research Engineer, Bangkok, Thailand

Jan 2025 to Aug 2025

- Developed a dual-mode assistive interface using a standard webcam (eye-tracking cursor control and dwell-click selection), achieving 40% higher precision than standard webcam-based gaze methods
- Deployed the system for active use by 50+ students with mobility disabilities; the school has since expanded to 100+ daily users and partnered with Tobii for further development
- Conducted formal user research with 200+ students and teachers, comparing eye tracking against switch controls and head tracking; findings shaped the final input model and usability design
- Built real-time gaze estimation and facial landmark tracking using OpenCV and MediaPipe; wrote full technical documentation covering implementation, usability metrics, and deployment

WPI Computer Science Department - Peer Learning Assistant, Worcester MA

Aug 2024 to May 2026

- Supported 300+ students across CS courses through weekly office hours, grading feedback, and lab sessions
- Courses: Database Systems II, Machine Learning, Operating Systems, Algorithms, Object-Oriented Design

Projects

Exercise Form Classification System, Deep Learning Course

Jan 2026 - May 2026

- Built a two-stage pipeline combining YOLOv11 pose estimation (17 COCO keypoints per frame) with four downstream neural networks (BiLSTM, Temporal Transformer, TSM, and Skeleton CNN)
- EfficientNet-B0 fine-tuned achieved 84.4% exercise classification accuracy and 81.9% form correctness accuracy across 168 exercise classes
- Processed 289,000 video clips from the Qualcomm Exercise Video Dataset, extracting and normalizing keypoint arrays to reduce a 100GB+ raw dataset to lightweight (30, 17, 3) input tensors; evaluated all models on accuracy, precision, recall, F1, and AUC-ROC
- Stack: Python, TensorFlow, PyTorch, YOLOv11, BiLSTM, Transformer, TSM, EfficientNet-B0

Brigham and Women's Hospital Kiosk Application, Software Engineering Course

Mar 2024 - May 2024

- Led a team of 10 engineers to deliver a full-stack hospital kiosk with 10+ features following AGILE methodology
- Stack: React, Node.js, Golang, PostgreSQL, Docker, AWS EC2, Auth0, CI/CD

Skills

AI/ML: LLM integration, YOLO, LSTM, RAG pipelines, prompt engineering, vector embeddings, sentence-transformers

Languages: JavaScript, TypeScript, Python, Java, C/C++, C#, HTML/CSS

Frameworks: React.js, Node.js, Express.js, Next.js, TensorFlow, PyTorch, OpenCV, MediaPipe

Databases: MongoDB, PostgreSQL, Firebase, Supabase, Prisma

Tools: Git, Docker, AWS (EC2, RDS), Auth0, ffmpeg, Tesseract, Whisper